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Preface

Well, several years have gone by and, happily, I've gotten requests for a second edition. So here it is. Since I now get another chance to wax eloquent, I'm going to make the most of it.

Those who're new to the book are possibly wondering why it exists, i.e., what makes it any different from "the competition." My claim to fame, if I have one, is that I write "to the students" as opposed to writing for the instructors using the book. These really are different audiences and I believe that the approach I've chosen makes a significant difference in the interest/motivation and ultimate learning that takes place. By "writing for the instructor" I mean writing a long and detailed text that strives for completeness and rigor, something that an expert in the field will appreciate and value. The problem is that at this point in their careers, students encountering the book are just beginning to learn vibrations. Just as a dance instructor would never overwhelm a beginning ballerina with complicated pirouettes and arabesques, there's nothing to be gained by piling vibrations esoterica onto someone at the start of his or her studies.

The approach I take is to introduce the topics of interest in a physically motivated way and to then show how one can use the material in a useful manner. Once that's done I go into some of the underlying details. The tone of the text is decidedly familiar. Many students have e-mailed and told me that it seemed that I was there with them, explaining the material to them in a one-on-one sort of way, as they read the book. That's exactly what I was aiming for. One student even went so far as to say the book makes good bedtime reading, but I think they might have been going a little overboard on that observation.

All of us that've gone through years of wading through textbooks know that many of them are boring to the point of tears. It's just not helpful, in my opinion, to have a fat book that's crammed with facts but is so inaccessible that the student doesn't get any motivation to read it. So I've chosen to

limit the number of topics I present and to present them in as engaging a manner as I can manage. As an aside, it's been my experience that nobody can actually cover all the material in my book, let alone in some of the more densely packed chapters, in a typical semester or one- or two-quarter course. Thus, stuffing more material merely serves to make the book more difficult to lug around and more costly. I'd love to think that the majority of the students who use the book subsequently keep it close to their hearts, but the sad reality is that after a course is finished, most all textbooks find new life as used books in the school bookstore. Thus pumping the book full of facts so that it can serve as a lifetime reference becomes somewhat problematic.

I use MATLAB in the book and highly recommend that anyone pursuing a degree in engineering go out and buy a student copy of MATLAB. It'll help in doing the problems in this book and it'll help in most of the person's other courses. Or, check if your school has MATLAB already installed on school machines. When I was first learning vibrations I had to take a lot of the material on faith, since it was too much of a chore to verify some of the assertions through calculation. Having a convenient software package running on a PC or a larger machine makes it easy to check the book and also easy to try different cases—to poke around with the mathematics a bit.

It was a temptation to include a disk with presolved problems that students could fire up at will. But, as you can see, I didn't. There were a couple of reasons. First, some instructors have no interest in MATLAB and some schools don't have it resident on their machines. Thus, they'd be subsidizing the software for schools that do use it. More importantly, I feel that MATLAB is very user friendly and that the more time you put into it, the more you'll be inclined to use it and to learn more about it. None of the problems in the book really require more than a few lines of commands to run, and writing them is fairly quick and painfree. Thus I've indicated what commands will enable the reader to solve the problems in the book, and I've included some explicit MATLAB commands so that readers planning to use this package can get started.

The material in the book grew out of lectures to junior, seniors, and first-year graduate students at Georgia Tech and at Berkeley. My presumption is that the student using the text has encountered linear algebra as a sophomore (but has likely forgotten much of it), has had a course in deformable bodies, and has taken a first course in dynamics. The text starts off with the classical material (single-degree-of-freedom systems) and then moves into more modern material. A good deal of emphasis is given to multiple-degree-of-freedom systems, as this subject is probably the most important for the practicing engineer. Whether obtained through a modal approach or a lumped discretization, most practical vibrations problems ultimately lead to multiple-degree-of-freedom systems.

Many of the problems are meant to challenge the student to think deeply about a particular problem, to really analyze why a certain thing happens and what can be accomplished with a given technique. The instructor should definitely look at the solution book before assigning a problem, as some have very short solutions whereas others are much more involved. Many absolutely require the use of a computer and a good set of eigenanalysis algorithms.

One unique aspect of the book is Chapter 7, Seat-of-the-Pants Engineering. There are many ways to get a quick answer or to verify an analysis that are known to experienced engineers but are picked up only slowly, after years of working on vibration problems. And yet these approaches

are useful for everyone to know right from the start. This chapter brings together a few of these approaches and applies them to all the systems that have been discussed. I felt that this material deserved the recognition of a separate chapter because it essentially carries the idea of approximate analyses (discussed in Chapter 6) to another level. Rather than being an approximate analysis that, with more effort, approaches the actual solution, these techniques are inherently approximate approaches that can't be made to emulate the real solution but can give the user a fast and accurate estimation of the real solution.

I hope that this book strikes the right balance between detail and accessibility and that it makes learning vibrations just a bit more fun than might otherwise be the case. I'd really enjoy hearing from you and finding out if the book achieved its aims. If you want to pass on your compliments or criticisms, I can be reached at bhtongue@newton.me.berkeley.edu. Errata, updates, and answers to selected problems can be found at

http://www.me.berkeley.edu/faculty/tongue/prinvib4.html.

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This book wouldn't exist without contributions from many people. My views on the whole field of dynamics were molded by my Ph.D. advisor, Earl H. Dowell, a masterful researcher whose abilities span an impressive range and whose ability to see the simple idea that underlies a confusingly complex phenomenon is much admired (by me) and but poorly approximated.

Many students have given me the benefit of their reactions to this material, both in lecture notes and in book form, and their input has led to a better book. To everyone that pointed out something that could be improved—thanks.

Since the first edition of this book appeared, I've received feedback from numerous people, all of which was appreciated. One of these correspondents, Colonel Wayne E. Whiteman, of the US Military Academy, truly deserves special mention. Colonel Whiteman went over my book with a fine-tooth comb and e-mailed me on many, many occasions to point out areas that were unclear, ambiguous or, most embarrasingly, incorrect as stated. His contributions helped to markedly improve the book in its second printing. My fond hope is that he'll have nothing to write to me about in this second edition.

Thanks to my son Curtis for transcribing some of my sloppily written problems into electronic form.

Peter Gordon and Karen Shapiro, editors at Oxford University Press, were there to help guide the book along into finished form and their help was greatly appreciated.

And finally, I'd like to thank Brenda Griffing for her marvelously complete and truly excellent job of copy editing. She found typos that I certainly wouldn't have noticed and showed a deft hand at sentence management that kept my writing style intact while simultaneously increasing clarity.